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Nickel (Ni) is a constituent of urease and hydrogenase.

c. **Osmotic Potential:** Potassium (K) plays a key role in maintaining osmotic potential of the cell. The absorption of water, movement of stomata and turgidity are due to osmotic potential.

d. **Energy components**: Magnesium (Mg) in chlorophyll and phosphorous (P) in ATP.

12.2 Functions, mode of absorption and deficiency symptoms of macronutrients

Macronutrients, their functions, their mode of absorption, deficiency symptoms and deficiency diseases are discussed here:

1. **Nitrogen (N):** It is required by the plants in greatest amount. It is an essential component of proteins, nucleic acids, amino acids, vitamins, hormones, alkaloids, chlorophyll and cytochrome. It is absorbed by the plants as nitrates (NO_3).

   Deficiency symptoms: Chlorosis, stunted growth, anthocyanin formation.

2. **Phosphorus (P):** Constituent of cell membrane, proteins, nucleic acids, ATP, NADP, phytin and sugar phosphate. It is absorbed as $H_2PO_4^+$ and HPO_4^- ions.

   Deficiency symptoms: Stunted growth, anthocyanin formation, necrosis, inhibition of cambial activity, affect root growth and fruit ripening.

3. **Potassium (K):** Maintains turgidity and osmotic potential of the cell, opening and closure of stomata, phloem translocation, stimulate activity of enzymes, anion and cation balance by ion-exchange. It is absorbed as K^+ ions.

   Deficiency symptoms: Marginal chlorosis, necrosis, low cambial activity, loss of apical dominance, lodging in cereals and curled leaf margin.

4. **Calcium (Ca):** It is involved in synthesis of calcium pectate in middle lamella, mitotic spindle formation, mitotic cell division, permeability of cell membrane, lipid metabolism, activation of phospholipase, ATPase, amylase and activator of adenyl kinase. It is absorbed as Ca^{2+} exchangeable ions.

   Deficiency symptoms: Chlorosis, necrosis, stunted growth, premature fall of leaves and flowers, inhibit seed formation, Black heart of Celery, Hooked leaf tip in Sugar beet, *Musa* and Tomato.

5. **Magnesium (Mg):** It is a constituent of chlorophyll, activator of enzymes of carbohydrate metabolism (RUBP Carboxylase and PEP Carboxylase) and involved in the synthesis of DNA and RNA. It is essential for binding of ribosomal sub units. It is absorbed as Mg^{2+} ions.

   Deficiency symptoms: Inter veinal chlorosis, necrosis, anthocyanin (purple) formation and Sand drown of tobacco.

6. **Sulphur (S):** Essential component of amino acids like cystine, cysteine and methionine, constituent of coenzyme A, Vitamins like biotin and thiamine, constituent of proteins and ferredoxin.plants utilise sulphur as sulphate (SO_4^-) ions.